« Artificial Intelligence, the Modern Frankenstein » Hybridity and Plurality in Technological Advancement

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-
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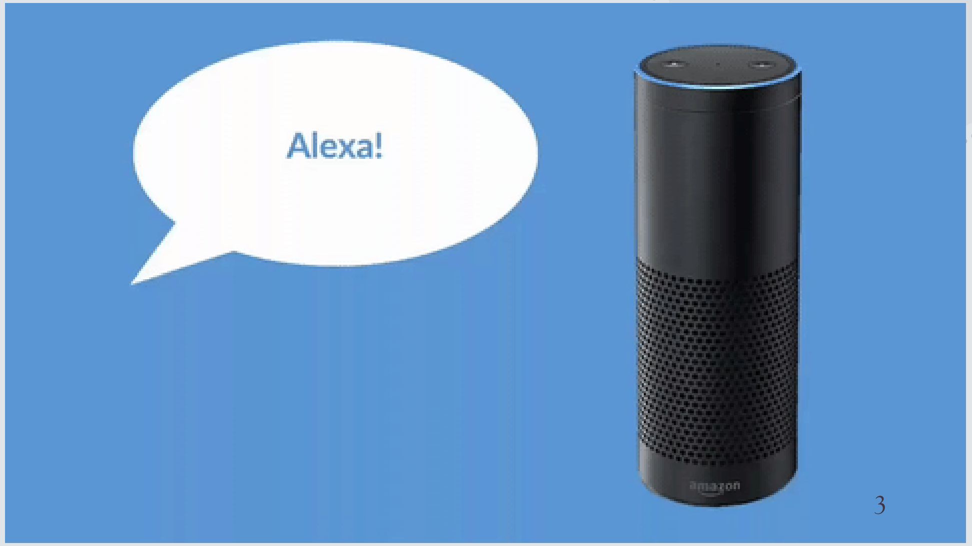
Overview

- What is Computational Linguistics?
- "Artificial Intelligence, the Modern Frankenstein"
- How can we use Computational linguistics to develop practical tools?
- Introduce my own research

Computational Linguistics – Definition

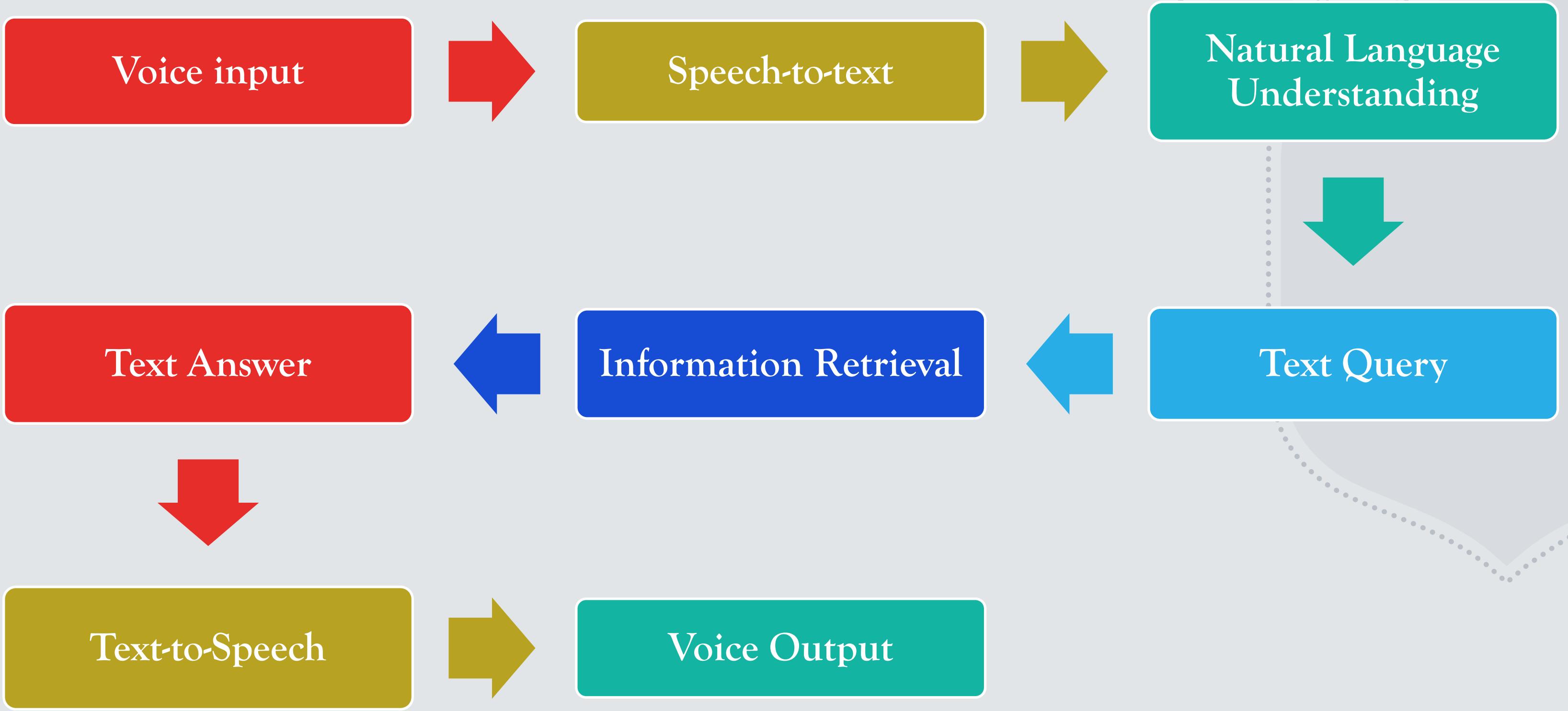
- The use of computer science technology to the analysis, comprehension and generation of written and spoken language.
- At the intersection of linguistics, computer science, artificial intelligence, math and logic, cognitive science/psychology, etc.
- Extremely versatile field of research with an emphasis on interdisciplinary research

- Today's focus : What are the different components to build a Siri/Alexa?



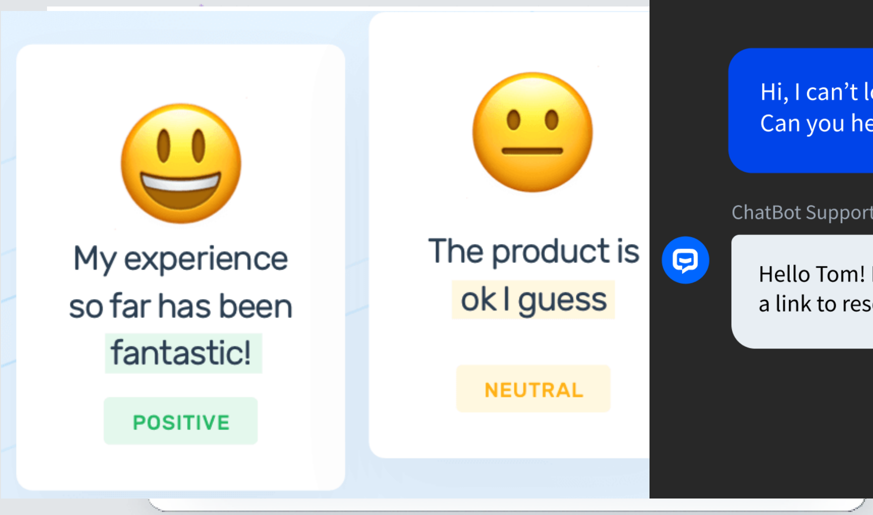
Plurality in Artificial Intelligence

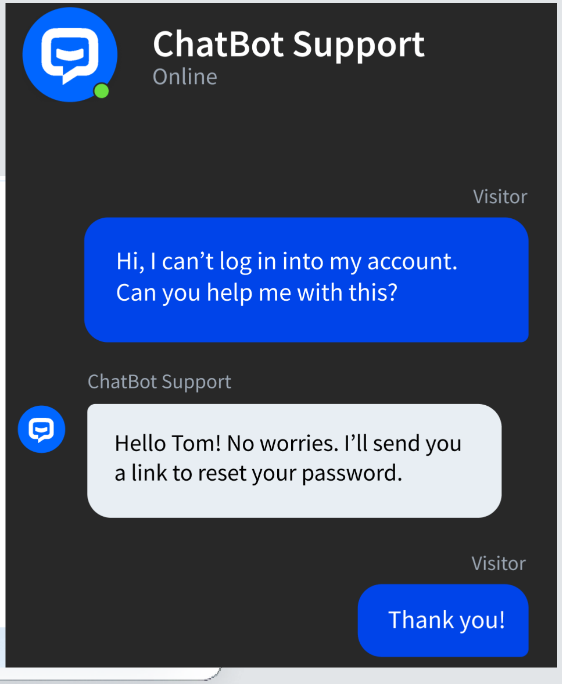
- Alexa, Siri, Google Assistant, ChatGPT are "artificial intelligence" which are as good as the technology used for each of the following steps.



Hybridity in Artificial Intelligence

- Main domains of linguistics :
  Morphology
  Syntax
  Semantics / Pragmatics
  Phonology / Phonetics





Computational Linguistics' contributions to AI technology:

- Grammar/Spell checker, Word prediction
- Automatic translators
- Language detection
- Sentiment analysis
- Chatbox
- Speech recognition
- Document indexing
- Text summarization/generation
- Anything language-related

Artificial Intelligence: The Modern Frankenstein

- **Artificial Intelligence Knowledge**

  AI systems know only as much as the data we provide them with

  - It cannot know what it hasn't been taught → Motivates developing NLP/CL research in Creoles

  An AI model is only as good as its inner architecture and mostly the quality of the data

  - It cannot make good data from bad data except if assisted → Motivates improving pre-existing datasets and providing deeper linguistic analysis

- **Human-like reasoning**

  An AI model does not "reason" the same way humans do (think of parrots)

  - It creates and establishes meaning of words based off of their linguistic context only → Motivates providing deeper semantic and reasoning depth to datasets

Morphology

- The study of word composition from the smallest unit that bears meaning (morpheme)
- Analyze the morphological composition of a given word and extract information such as:

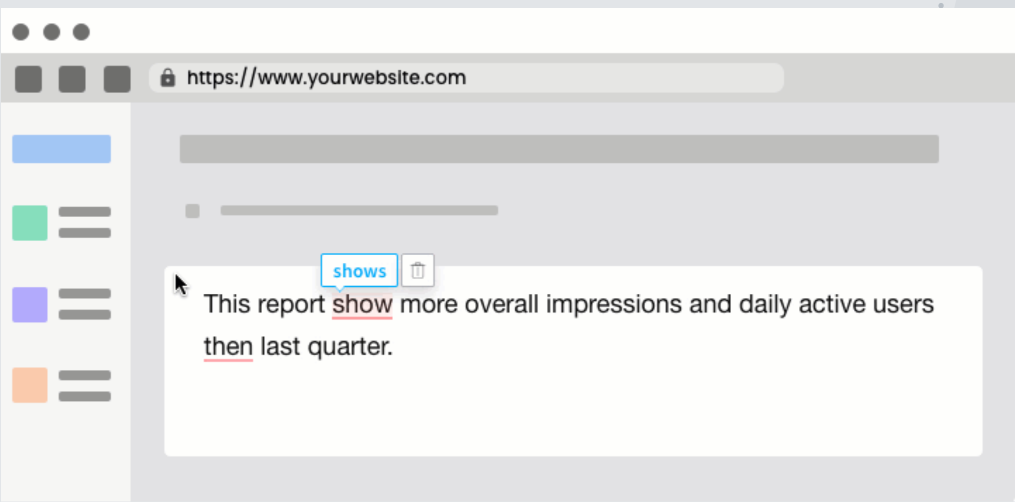
Lemma/Part of speech / gender / number / affixes/ inflection (verbs)

The	dog	chases	the	cats
the +DET +DEF	dog +N +SG	chase.s +V +3SG +PRES	the +DET + DEF	cat +N +PL

The dog chases the cats.

How does morphology contribute to AI technology? Why is morphological information important in NLP?

Grammar checker



Bag of words
cat
chases
dog
the
A
Cats
Dogs
Chase

The dog chases a cat
The dogs chase the cats

The dog ***chase*** *the cat.*
A ***dogs*** *chase the cats.*
The dogs ***chases*** *the cats.*

Noun-Verb agreement
Determiner-noun agreement
→ gender, number, pos

Bag of words
Cat = N 3p. Sing.
Chases = V 3p. Sing.
Dog = N 3p. Sing
The = DT Sing./Plur.
A = DT Sing.
Cats = N 3p. Plur.
Dogs = N 3p. Plur.
Chase = V 3p. Plur.

Grammar checker

Bag of words
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A = DT Sing.
Cats = N 3p. Plur.
Dogs = N 3p. Plur.
Chase = V 3p. Plur.
Yesterday = Adv

→ *The dog chases a cat yesterday* →

Bag of words
Cat = N 3p. Sing.
Chase.s = V 3p. Sing. **Prs.**
Dog = N 3p. Sing
The = DT Sing./Plur.
A = DT Sing.
Cats = N 3p. Plur.
Dogs = N 3p. Plur.
Chase = V 3p.Plur. **Prs.**
Yesterday = Adv. **Pst.**

Tense agreement
→ inflection/temporal info

Grammar checker

Bag of words
Cat = N 3p. Sing.
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A = DT Sing.
Cats = N 3p. Plur.
Dogs = N 3p. Plur.
Chase = V 3p. Plur. Prs.
Yesterday = Adv. Pst.
Chase.d = V Pst.

The dog chased a cat yesterday

=

The dog chased the cats yesterday

=

The dogs chase the cat

=

A dog chases a cat

→Lemmatization

Dictionary
CAT Cat = N 3p. Sing. Cats = N 3p. Plur.
CHASE Chases = V 3p. Sing. Prs. Chase = V 3p. Plur. Prs. Chased = V Pst.
DOG Dog = N 3p. Sing Dogs = N 3p. Plur.
THE The = DT Sing. /Plur.
A A/AN = DT Sing.
Yesterday = Adv. Pst.

Morphology

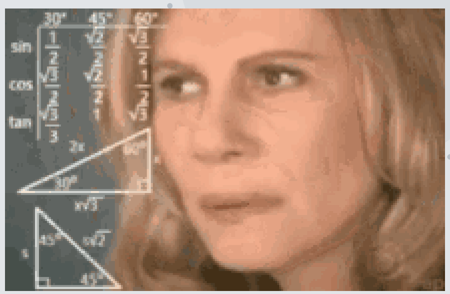
- Building a morphological analyzer/grammar checker:

Rule-based approach = only as good as the rules you create and the dictionary you have

Dictionary of roots, affixes, inflections, and grammar of transformation rules

The the +DET +DEF	dog dog +N +SG	chases chase.s +V +3SG +PRES	the the +DET + DEF	cats cat +N +PL

The the +DET +DEF	math math +N +SG	is be+V +3SG +PRES	mathing math +V +PROG



Statistical/neural approach = only as good as the corpus and the amount of data you provide

Annotated or unannotated data, probability that the unlabelled word is the same as other words found in a similar context from the dataset

Probabilities calculated based on all the available information/features (word order, pos, tense, etc.).

Event Sequencing Annotation with TIE-ML

Damir Cavar, Ali Aljubailan, Ludovic Mompelat, Yuna Won, Billy Dickson, Matthew Fort, Andrew Davis, and Soyoung Kim. 2022.

- Some event sequencing processes are universal, and/or dictated by the structure of a given clause. These clues are detected by humans, but not so much by AI models.
- Event sequencing annotation :

Wash the veggies,	chop them,	and fry them
1	2	3

Before you fry the veggies,	wash them,	and chop them
3	1	2

- Morphological Information can help provide a model enough features to make its own connections

Event Sequencing Annotation with TIE-ML

Damir Cavar, Ali Aljubailan, Ludovic Mompelat, Yuna Won, Billy Dickson, Matthew Fort, Andrew Davis, and Soyoung Kim. 2022.

John **said** that he **left** early

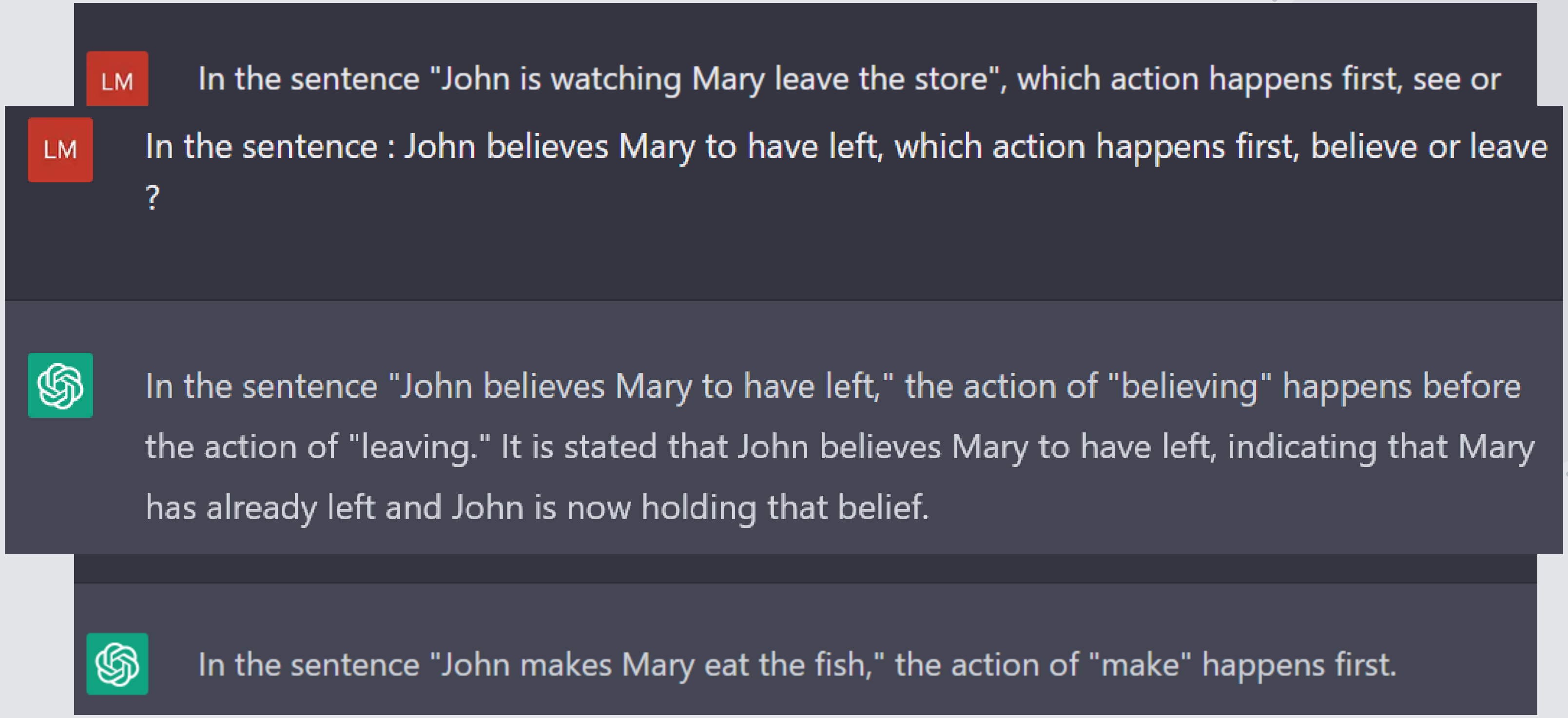
2 1

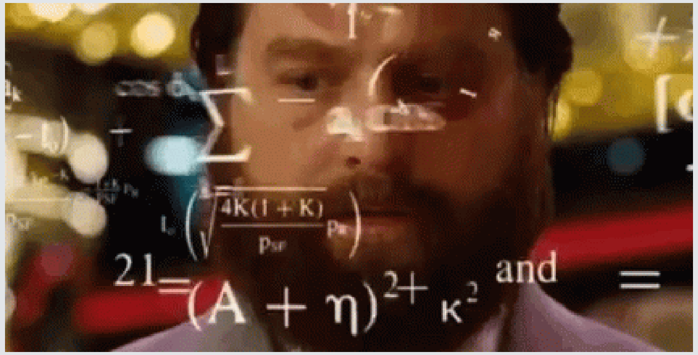
John **believes** Mary to **have left**

2 1

John is **watching** Mary **leave** the store

John **makes** Mary **eat** the fish





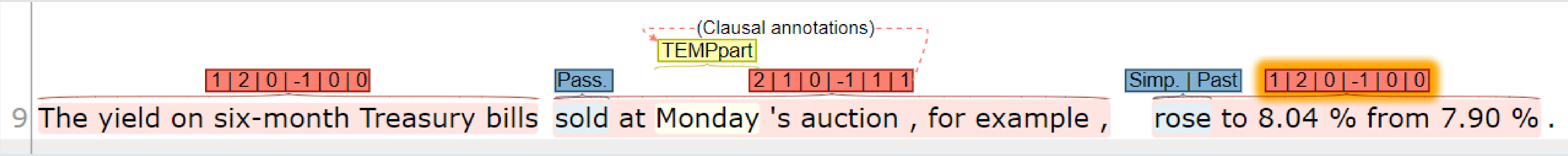
Event Sequencing Annotation with TIE-ML

Damir Cavar, Ali Aljubailan, Ludovic Mompelat, Yuna Won, Billy Dickson, Matthew Fort, Andrew Davis, and Soyoung Kim. 2022.

A clause must have one and only one predicate. Each predicate is marked for tense and aspect.

- **Clause ID** : linear order of a given clause
- **Time Slot** : chronological order of a given clause
- **Speaker, Event times** : Reichenbach's theory of tense
- **Level** : Independent/main clause (=0), subordinate clause (=level of main clause -1)
- **Selected by** : Clause ID of the dominating clause

Annotation schema
Clause ID
Time Slot
Speaker time
Event time
Level
Selected by



Dictionary
CAT Cat = N 3p. Sing. Cats = N 3p. Plur.
CHASE Chases = V 3p. Sing. Prs. Chase = V –[3p. Sing.] Prs. Chased = V Pst.
DOG Dog = N 3p. Sing Dogs = N 3p. Plur.
THE The = DT Sing. /Plur.
A A/AN = DT Sing.
Yesterday = Adv. Pst.

The dog chased a cat yesterday
The dogs chased the cats yesterday
The dogs chase the cat

Dogs the cat chase
The dogs chased a yesterday cat

Word order rules
Constituent rules

Syntax

Syntax

- The study of composition of sentences and the relationship between their components
- Syntactic parsing = automatic analysis of the syntactic structure of languages
- An important part of computational linguistics

  Constituency parsing & Dependency parsing: provide more information on the sentence and its components

Constituency parsing

Det + Noun + Verb + Det + Noun + Adv

Det + Noun + Verb + Det + Noun

Noun + Verb + Noun

The dog chased a cat yesterday

The dogs chased the cat

Dogs chase cats

Constituency rules

Dogs the cat chase

The dogs chased a yesterday cat

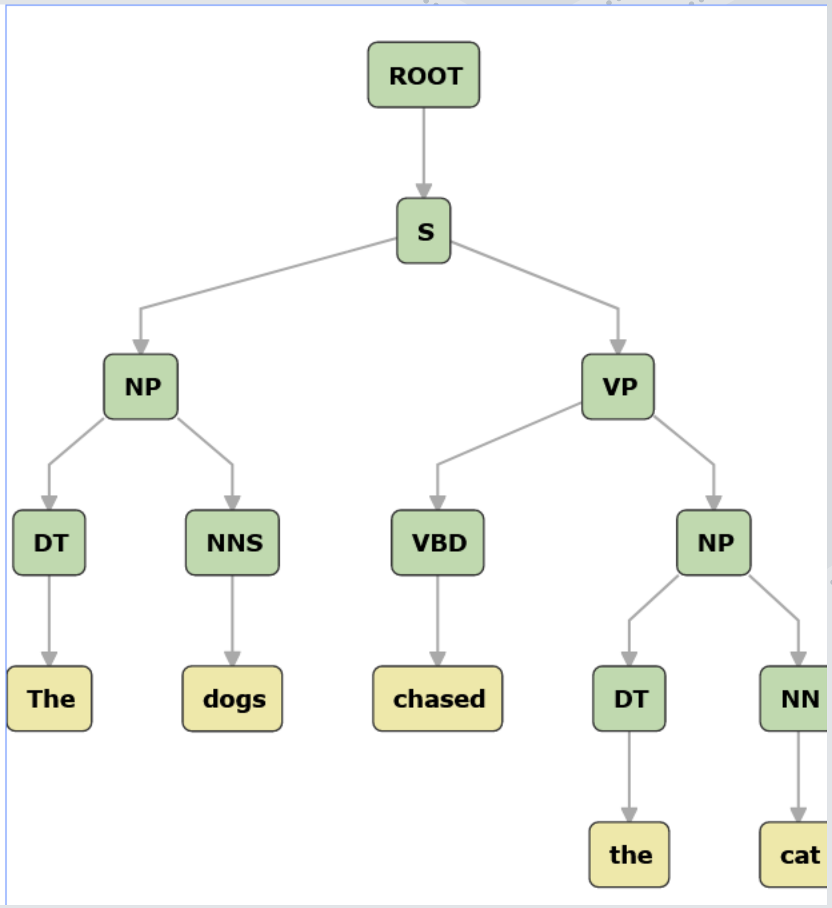
NP : [Det + Noun | Noun]

VP : [Verb + NP | Verb]

AdvP : [Adv]

S : [NP + VP (+Adv)]

https://corenlp.run/



Dependency parsing

The dog ate an apple yesterday
The dogs ate the apple
Dogs eat apples

Apples eat dogs
The apple eats the dog
An apple ate a dog

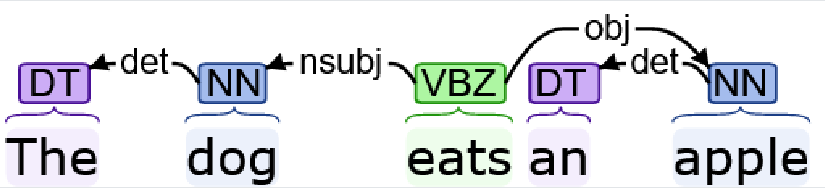
NP : Det + Noun | Noun

VP : Verb + NP | Verb

AdvP : Adv

S : NP + VP (+Adv)

Importance of dependency relations



https://corenlp.run/

Subject vs. object

~~*Eat : Subj = apple | Obj = dog~~

Eat : Subj = dog | Obj = apple

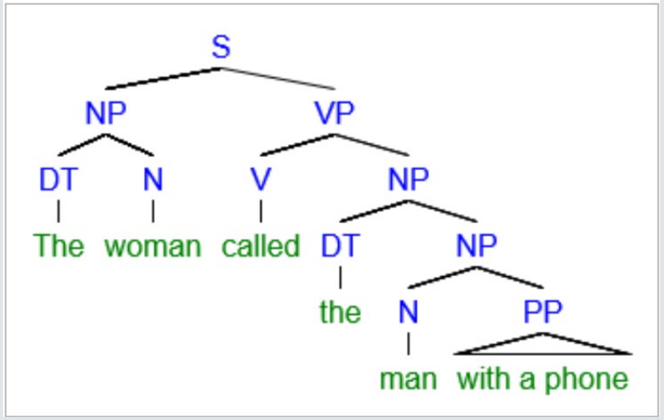
Syntax

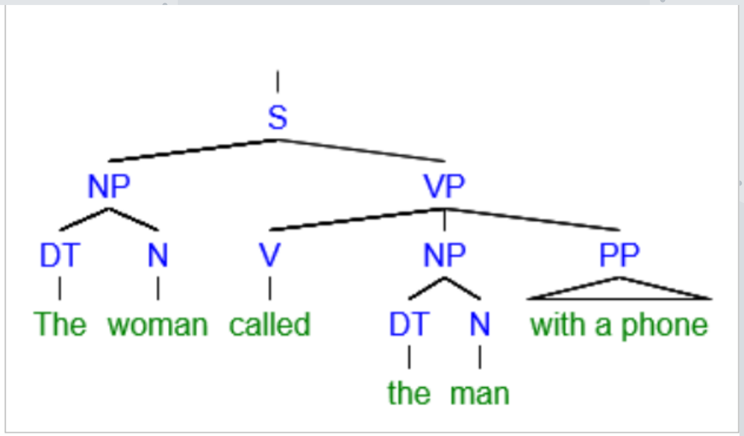
Disambiguating sentences :

"The woman called the man with a phone"

[The woman] [called [the man with a phone]

[The woman] [called [the man] with a phone]





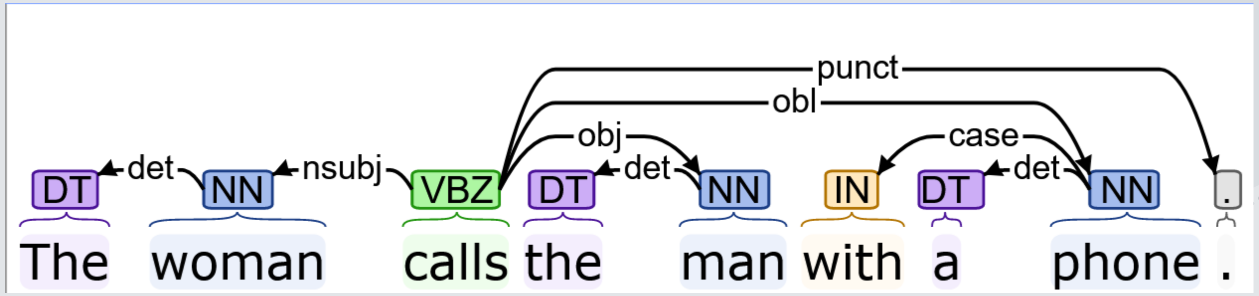
Syntax/Semantics interface :

Useful for semantic information extraction.

What is the action? Who is the agent? Who is undergoing the action?

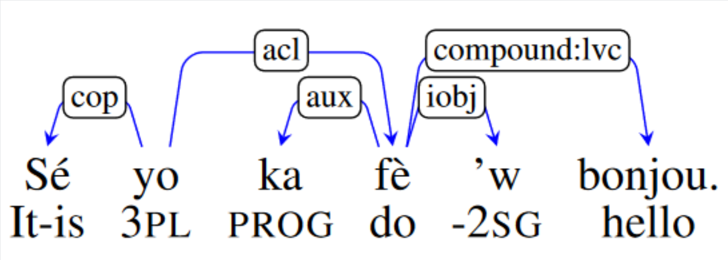
[[AGENT: woman]; [EVENT: call]; [PATIENT: man with a phone]]

[[AGENT: woman]; [EVENT: call with a phone]; [PATIENT: man]]



How to Parse a Creole: When Martinican Creole Meets French

Ludovic Mompelat, Daniel Dakota, and Sandra Kübler. 2022.



- "They are the ones greeting you"
- Automatic parser of Martinican Creole

Neural-based parsers require a large amount of data/big language models

The more data the better the training of a model

1 annotator – 240 MC sentences annotated for POS and dependencies

French corpus of 16,000 sentences to leverage

Concatenation/Transfer learning/ Multi-task learning to compensate

Concatenation of datasets was most efficient (80,75 UAS%, 71,77% LAS)

Token	POS
Sé	VERB
yo	PRON
ka	PART
fè	VERB
'	PUNCT
w	PRON
bonjou	NOUN
.	PUNCT

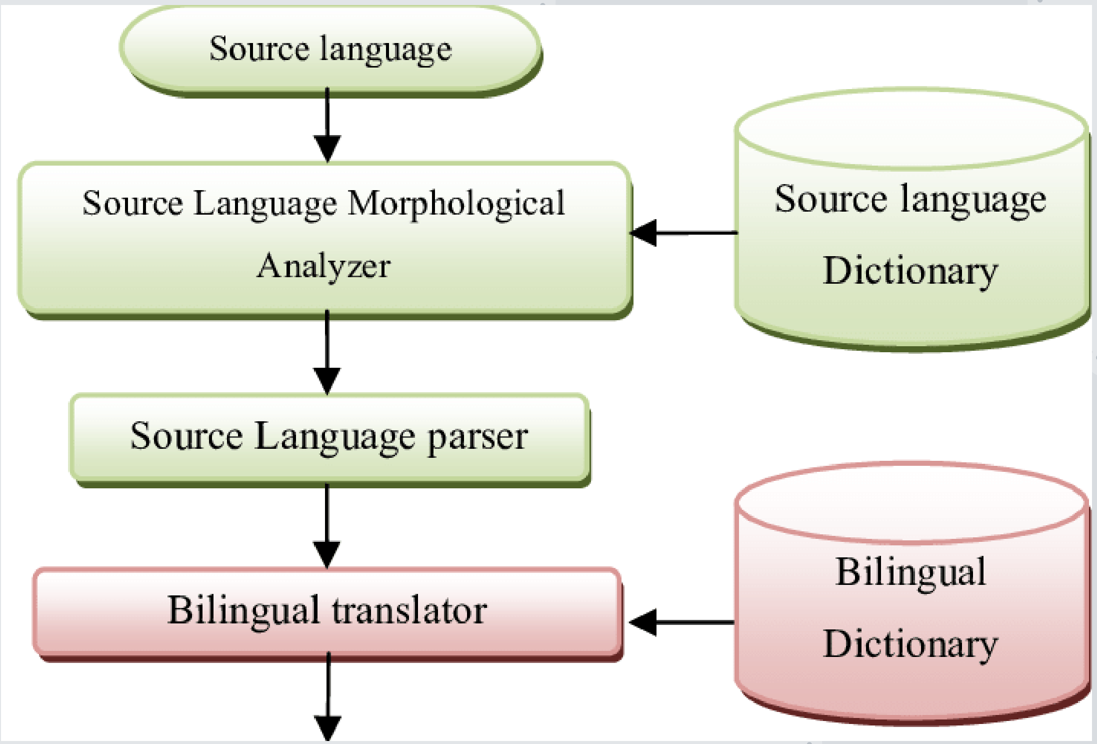
	UAS	LAS
	25.05	11.73
	65.08	51.95
	38.23	21.63
	62.89	48.36
	71.71	62.86
	63.36	49.83
	72.95	60.57
	80.75	71.77
BERT	72.17	58.57

Table 2: Baselines for training on French, Martinican (MC), and concatenated French+Martinican (FR+MC).

Machine Translation

How to translate “The cat chases the mice” into French?

The the +DET +DEF	cat cat +N +SG	chases chase.s +V +3SG +PRES	the the +DET + DEF	mice mouse +N +PL

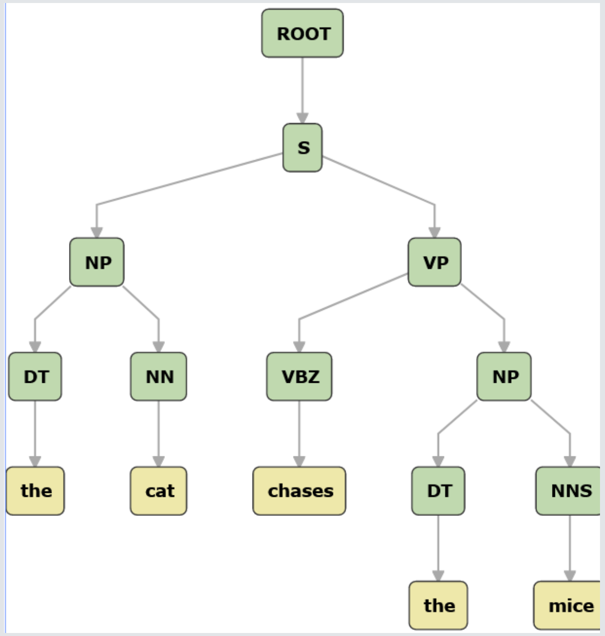


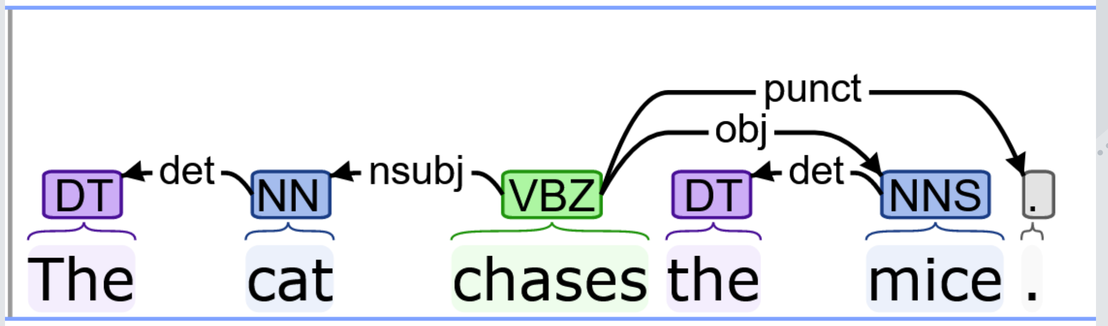
NP : Det + Noun | Noun

VP : Verb + NP | Verb

AdvP : Adv

S : NP + VP (+Adv)

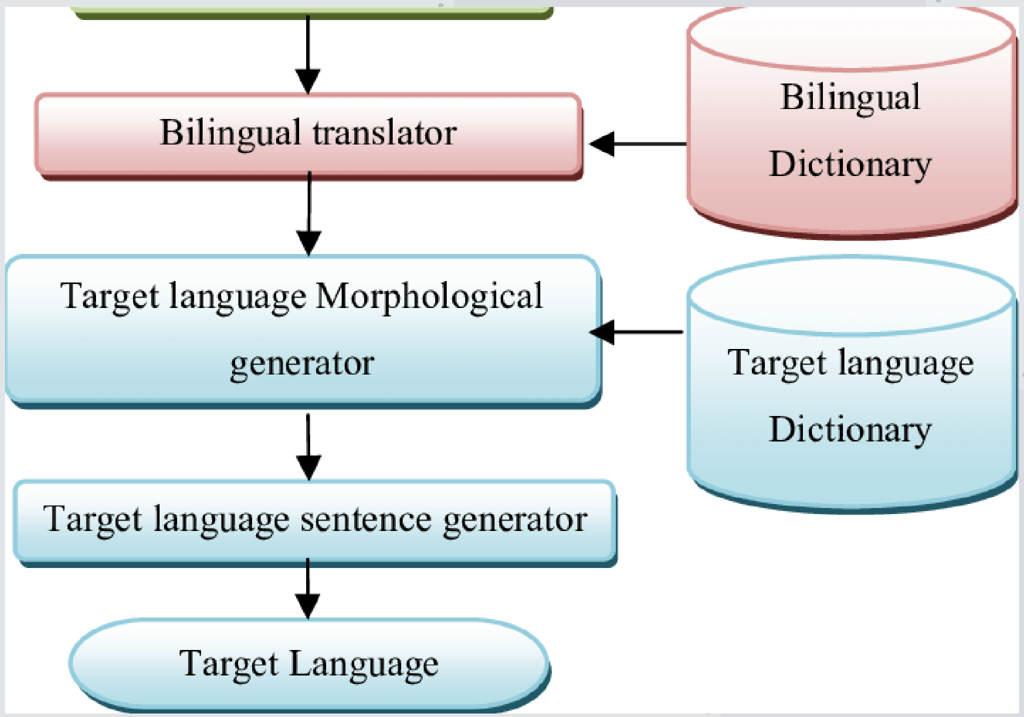




Machine Translation

The cat chases the mice

English	French
CAT Cat = N 3p. Sing. Cats = N 3p. Plur.	CHAT Chat = N 3p. Sing. Masc. Chats = N 3p. Plur. Masc.
CHASE Chases = V 3p. Sing. Prs. Chase = V –[3p. Sing.] Prs. Chased = V Pst.	CHASSER Chasse = V 3p. Sing. Prs. Chassent = V 3p. Plur. Prs. Chassait = V 3p. Sing Pst.
MOUSE Mouse = N 3p. Sing Mice = N 3p. Plur.	SOURIS Souris = N 3p. Sing./Plur. Fem.
THE The = DT Sing. /Plur.	LE/LA/LES Le = DT Sing. Masc. La = DT Sing. Fem. Les = DT Plur. Masc.



+

NP : Det + Noun | Noun

VP : Verb + NP | Verb

AdvP : Adv

S : NP + VP (+Adv)

=

"Le chat chasse les souris"

Machine Translation

The **blue cat** chases the **nice mouse** :: Le **chat bleu** chasse la **gentille souris**

Different word order

He **is** talk**ing about** the **weather** :: Il **parle du temps**

English	French
Be + V-ing	Present simple
About	Sur, à propos de, environ, sur le point de,
Weather, time, tense	Temps

- Statistical machine translation :

Very large annotated bilingual corpus, find the closest analogue of the target segment.

Semantics/pragmatics

Difficult field to apply to AI

A lot of what we understand is not made explicit but is easily inferred by humans

- **Sentiment Analysis**

“I did not really like this product. I would not suggest buying it”

Syntactic and lexical information

- **Metonymy**

“**The White House** wrote a press release”

Syntactic information/Ontology

- **Anaphora resolution – Inference**

“Take a **lemon** and a **knife**. Now cut **it** in half.”

- What is being cut?

Syntactic information / semantic roles

WordNet Search - 3.1
- WordNet home page - Glossary - Help

Word to search for: White House | Search WordNet

Display Options: (Select option to change) | Change

Key: "S:" = Show Synset (semantic) relations, "W:" = Show Word (lexical) relations

Display options for sense: (gloss) "an example sentence"

Noun

- S: (n) **White House**, EXEC (the chief executive department of the United States government)
  - *direct hypernym* / *inherited hypernym* / *sister term*
    - S: (n) executive department (a federal department in the executive branch of the government of the United States)
- S: (n) **White House** (the government building that serves as the residence and office of the President of the United States)
  - *part meronym*
    - S: (n) Oval Office (the office of the President of the United States in the White House)
  - *part holonym*
    - S: (n) Washington, Washington D.C., American capital, capital of the United States (the capital of the United States in the District of Columbia and a tourist mecca; George Washington commissioned Charles L'Enfant to lay out the city in 1791)
  - *instance*
    - S: (n) residence (the official house or establishment of an important person (as a sovereign or president)) *"he refused to live in the governor's residence"*
    - S: (n) government building (a building that houses a branch of government)

Cutting

Definition:

An Agent cuts a Item into Pieces using an Instrument (which may or may not be expressed).

At the ceremony, the CEO CUT the red ribbon hanging across the main entrance into a glorious confetti.

Noun

- S: (n) **knife** (edge tool used as a cutting instrument; has a pointed blade with a sharp edge and a handle)

Semantics/pragmatics

- **Coreference resolution**

  "**Barack Obama** is the **44th President of the USA**. **Michelle Obama's husband** cannot run for president anymore."

  Syntactic information / World knowledge

- **Knowledge graphs/database, ontologies**

  Storing and representing relationship between people, things, events, etc.

  Ontology = General world knowledge

  Entity > Animate> Animal > Mammal > Human > Person > James

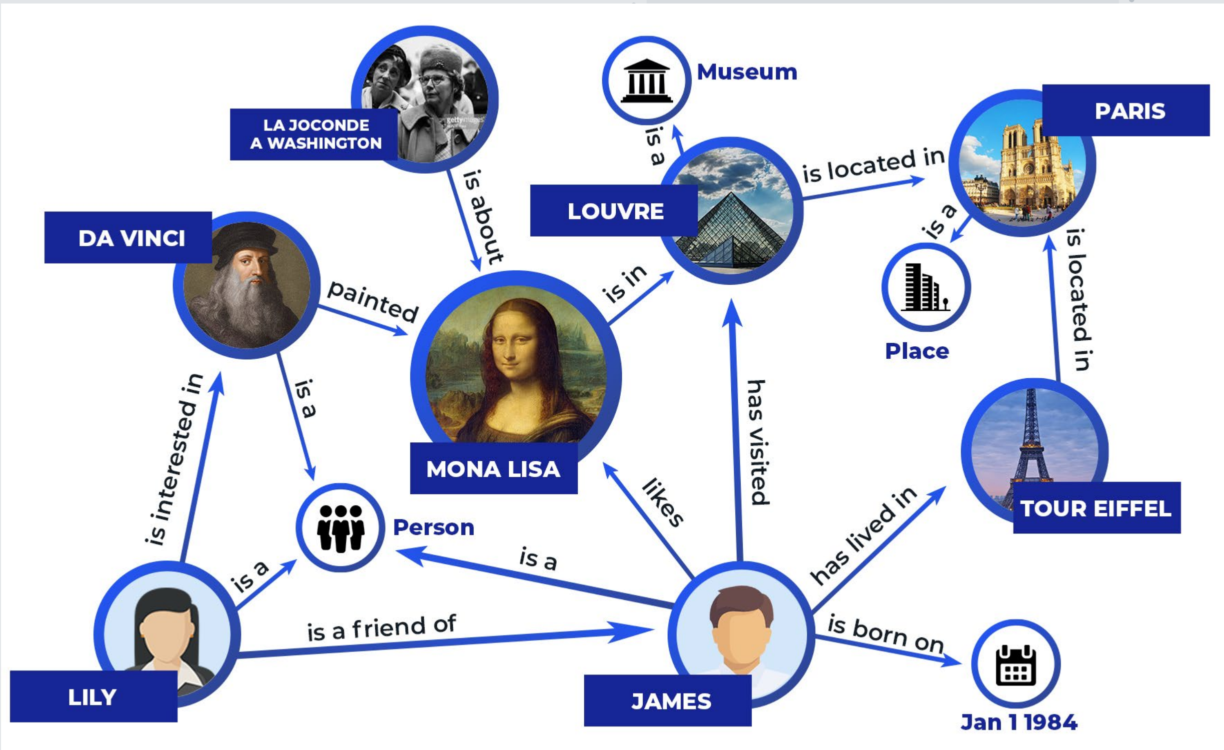
  Database = Specific knowledge

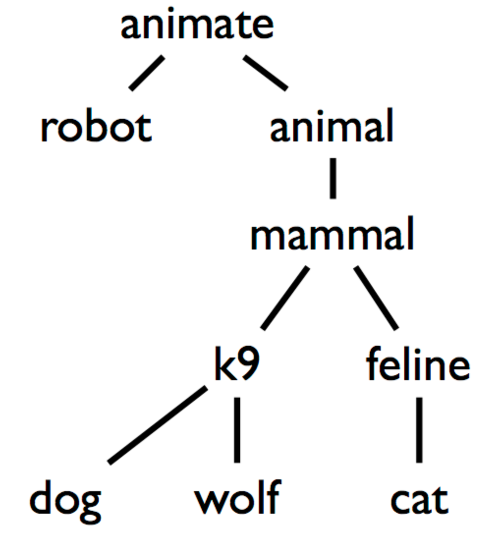
  Social media activity tracing/Weather forecast/trading

- **Named entity recognition (Information extraction):**

  "**Mary** went to **San Diego** for a **CL** conference last **May**"

  PERSON LOC SCIENCE TEMP





Hard-to-crack cases

Many AI tasks are **Classification tasks** – Two sets of datasets :

One with documents that have the targeted phenomenon

One with randomized documents

Language model trains and learns features that separates both sets and then classifies unseen docs

- **Hate Speech detection – Sentiment Analysis**

  "They should go back to their country"

- **Fake news detection – Syntactic analysis and database on news**

  "The last presidential elections were stolen"

- **Sarcasm detection – Context cues**

  "Thank you for always keeping the apartment clean"

- **Conspiracy Theory detection**

How “Loco” is the LOCO Corpus? Annotating the Language of Conspiracy Theories

Ludovic Mompelat, Zuoyu Tian, Amanda Kessler, Matthew Luettgen, Aaryana Rajanala, Sandra Kübler, and Michelle Seelig. 2022.

Creating an annotation schema to classify documents as containing Conspiracy Theory content/language/belief or not.

- **Content** = The theme of the document, does it talk about a conspiracy?

  Contradictory opinion to mainstream opinion / sensationalism / mention of other CTs

- **Language** = How is the document linguistically constructed?

  Use of caps/atypical named entities / unconventional punctuation!!!/Frequent use of 1st and 2nd person pronouns / Frequent questions directed at the reader / paraphrasing instead of direct quoting

- **Belief** = Does the author of the document believe and seek to propagate the conspiracy?

- “FBI *says* No One Killed at Sandy Hook” [LOCO ID: C005a9]
- “They reported Fake numbers that they made up & don’t even exist. WE WILL WIN AGAIN!” [LOCO ID: C00775]
- “The Obama White House” [LOCO ID: C00a2d]
- “‘Sleepy Joe’ makes another gaffe on his campaign trail” [LOCO ID: C0690d]
- “If you want more evidence of a government seeking control, look no further than the IRS scandal where the Obama administration was using the IRS to stop conservatives and religious groups from organizing opposition.” [LOCO ID: C00650]

Phonology and phonetics

- Phonology is the study of the patterns of sounds in a language and across languages (syllables, tones, intonation, etc.)
- Phonetics is the study of the production and perception of speech sounds
- **Used for speech recognition / speech-to-text**

  Formants, frequency, volume, word stress, pause, tones, etc. as features

  Speaker detection → Phoneme detection → Acoustic analysis → Language Modelling → Speech-to-text

Acoustic model = audio recordings of speech and their transcriptions compiled into statistical representations of the sounds for words

Language model = provides the probabilities of sequences of words.

How to build a personal assistant

- Alexa, Siri, Google Assistant are "artificial intelligence" which are as good as the technology used for each of the following steps.

Speech recognition
Phoneme detection

Voice input → **Speech-to-text**

Acoustic modeling
Language modeling

→ **Natural Language Understanding**

Syntactic Parsing
Information extraction

↓

Text Query

Text conversion to database query format

← **Information Retrieval**

Data extraction

← **Text Answer**

Sentence generation
Language modeling

↓

Text-to-Speech

Phonetic transcription generation
Acoustic modeling

→ **Voice Output**

Speech generation

Current research

Constrastive study of syntactic structure of complements in Martinican Creole and Haitian Creole.

Creation of the first annotated corpus, parser and POS tagger for these languages

- Stems from claims that there is no distinction between finite and non-finite(infinitive) clauses. Is there really no finite/non-finite distinction in these two creoles?
- Weigh in on the debate surrounding the linguistic concept of finiteness for creole languages, which may expand to other languages
- Looking at data documented in the literature from the past 40 years + data collected from newspapers, blogs, and native speakers. → Used for my corpus

Goal : Develop CL resources, tools and technology for under-resourced languages / Justify including creole languages in the most theoretical and the most practical aspects of linguistics and AI.

Thank you! ☺

For questions or discussions : lmompela@iu.edu

Some references to get started on Computational Linguistics

- **Textbook** : Speech and Language Processing (3rd ed.) Dan Jurafsky and James H. Martin https://web.stanford.edu/~jurafsky/slp3/
- **Cool Tools**

  Wordnet : http://wordnetweb.princeton.edu/perl/webwn

  Framenet : https://framenet.icsi.berkeley.edu/fndrupal/framenet_search

  Stanford Corenlp : https://corenlp.run/
- **Corpus Linguistics tools** :

  Concordancer https://www.lextutor.ca/

  Wordsmith https://wordsmith.org/

  Antconc https://www.laurenceanthony.net/software/antconc/

  British National Corpus (BNC) https://www.english-corpora.org/bnc/

  The Corpus of Contemporary American English (COCA) https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/AMUDUW